Initial Model Derivation: LCST Transport-Barrier Mechanism

User-provided preliminary derivation, organized for the repository

May 29, 2026

This file is the source of truth for long mathematical derivations. Markdown files should only summarize it.

Ingestion note. This document organizes the user-provided preliminary material from Sections II–III and the appendices of the attached manuscript source. The scientific content is kept at the candidate-model stage: equations and assumptions are recorded here, but no model, simulation result, or claim is validated by this repository.

1 Model variables

1.1 Physical system

The source model considers a planar thermoresponsive gel slab of reference half-thickness H_0 containing an immobilised catalyst for an exothermic, irreversible, first-order reaction. The slab is in contact at its free surface with a well-mixed bath of constant temperature T_∞ and constant reactant concentration c_b. The Lagrangian coordinate is

$$\xi = X/H_0 \in [0, 1], \tag{1}$$

with $\xi = 0$ the symmetry plane and $\xi = 1$ the free surface.

1.2 Kinematics and fields

In the dimensional appendix notation, the current position of a material point is $x(X, t)$ and the local swelling ratio is

$$J(X, t) = \frac{\partial x}{\partial X} > 0. \tag{2}$$

The polymer volume fraction is

$$\phi(X, t) = \frac{\phi_{p0}}{J(X, t)}, \tag{3}$$

where ϕ_{p0} is the polymer volume fraction in the reference state. The current slab thickness is

$$h(t) = x(H_0, t) = \int_0^{H_0} J(X, t)\, dX. \tag{4}$$

The three primary fields in the nondimensional model are

$$J(\xi, t) : \text{local swelling ratio}, \tag{5}$$

$$u(\xi, t) = c/c_b : \text{normalized reactant concentration}, \tag{6}$$

$$\theta(\xi, t) = (T - T_\infty)/\Delta T_* : \text{dimensionless temperature excess}, \tag{7}$$

where

$$\Delta T_* = \frac{(-\Delta H_r) c_b}{C_*} \tag{8}$$

is the adiabatic reaction temperature rise.

1.3 Dimensionless parameter table from the source model

Symbol	Physical meaning	Default	Range
Da	Reaction rate / swelling rate	4	3–18
Bi_μ	Surface permeation Biot number	1.0	–
Bi_c	Surface reactant Biot number	0.70	0.1–2
Bi_T	Thermal exchange number	0.10	0.05–0.5
S_χ	LCST sensitivity	1.0	0.5–8
Γ_A	Arrhenius sensitivity	1.5	0–7
D_0	Reactant diffusivity scale	2.0	0.5–3
m_{act}	Reaction accessibility exponent	6	1–6
m_{diff}	Diffusion tortuosity exponent	2	1–6
α	Thermal / swelling time ratio	0.03	–
δ	Diffusion / permeation ratio	0.08	–
ℓ	Interface regularization length	0.01	–
Ω_e	Elastic / mixing energy ratio	0.12	–
ϕ_{p0}	Reference polymer fraction	0.15	–
χ_∞	Interaction parameter at T_∞	0.60	–
χ_1	Concentration-dependent χ	1.10	–
ε_T	Nonlinear Arrhenius correction	0.03	–

2 Governing equations

2.1 Compact dimensionless PDE system

The compact nondimensional governing equations in the source manuscript are

$$\frac{\partial J}{\partial t} = \frac{\partial}{\partial \xi}\left[\mathcal{M}(J)\,\frac{\partial \mu}{\partial \xi}\right], \tag{9a}$$

$$\frac{\partial (uJ)}{\partial t} = -\frac{\partial}{\partial \xi}(u\,q) + \frac{\partial}{\partial \xi}\left[\mathcal{D}(J)\,\frac{\partial u}{\partial \xi}\right] - \mathrm{Da}\,J\,\widehat{R}, \tag{9b}$$

$$\mathcal{C}(J)\,\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial \xi}\left[\mathcal{K}(J)\,\frac{\partial \theta}{\partial \xi}\right] + \mathrm{Da}\,J\,\widehat{R}. \tag{9c}$$

Here

$$q = -\mathcal{M}\,\partial_\xi \mu \tag{10}$$

is the solvent flux, $\mathcal{M}$ is the swelling mobility, $\mathcal{D}$ and $\mathcal{K}$ are the effective reactant diffusivity and thermal conductivity, and $\mathcal{C}$ is the effective heat capacity. In this compact form, the prefactors δ and α are absorbed into $\mathcal{D}$ and $\mathcal{K}$. The enthalpy-advection term is neglected under the source assumption $\mathrm{Pe}_T \ll 1$.

2.2 Reaction rate

The compact source reaction rate is

$$\widehat{R} = u\,(1-\phi)^{m_{\mathrm{act}}}\,\exp\!\left(\frac{\Gamma_A\,\theta}{1+\varepsilon_T\,\theta}\right), \tag{11}$$

with $\phi = \phi_{p0}/J$. The appendix writes the general reaction order as

$$\widehat{R}(u,\theta,J) = u^n\,a(J)\,\exp\!\left[\frac{\Gamma_A\,\theta}{1+\varepsilon_T\,\theta}\right], \tag{12}$$

where

$$a(J) = \left(1-\frac{\phi_{p0}}{J}\right)^{m_{\mathrm{act}}} = (1-\phi)^{m_{\mathrm{act}}}. \tag{13}$$

The compact model corresponds to the first-order case $n = 1$.

2.3 Dimensional conservation laws from the appendix

The dimensional solvent conservation equation is

$$\frac{\partial J}{\partial t} = -\frac{\partial Q_s}{\partial X}, \tag{14}$$

with solvent flux

$$Q_s = -M_s(J,T)\,\frac{\partial \mu_s}{\partial X}. \tag{15}$$

The dimensional reactant balance is

$$\frac{\partial (Jc)}{\partial t} + \frac{\partial N_c}{\partial X} = -J\,R(c,T,J), \tag{16}$$

with total reactant flux

$$N_c = c\,Q_s - D_c(J,T)\,\frac{\partial c}{\partial X}. \tag{17}$$

The dimensional heat equation is

$$C_{\mathrm{eff}}(J)\,\frac{\partial T}{\partial t} = \frac{\partial}{\partial X}\left(K_T(J)\,\frac{\partial T}{\partial X}\right) - \frac{\partial}{\partial X}\left(c_p^{\ell}\,T\,Q_s\right) + (-\Delta H_r)\,J\,R(c,T,J). \tag{18}$$

2.4 Free energy and chemical potential

The Helmholtz free-energy density per unit reference volume is

$$\Psi(J,T) = \Psi_{\mathrm{el}}(J) + \Psi_{\mathrm{mix}}(\phi,T) + \frac{\kappa_J}{2}\left(\frac{\partial J}{\partial X}\right)^2, \tag{19}$$

where

$$\Psi_{\mathrm{el}}(J) = \frac{G}{2}\left(J^2 - 1 - 2\ln J\right) \tag{20}$$

and

$$\Psi_{\mathrm{mix}}(\phi,T) = \frac{R_g T}{v_s}\left[(1-\phi)\ln(1-\phi) + \chi(T,\phi)\,\phi(1-\phi)\right]. \tag{21}$$

The interaction parameter is

$$\chi(T, \phi) = \chi_0(T) + \chi_1 \phi. \tag{22}$$

The source appendix gives the mixing contribution to the solvent chemical potential as

$$\frac{\mu_{\mathrm{mix}}}{R_g T / v_s} = \ln(1 - \phi) + \phi + \chi(T, \phi)\, \phi^2, \tag{23}$$

and the elastic contribution as

$$\frac{\mu_{\mathrm{el}}}{R_g T / v_s} = \frac{G\, v_s}{R_g T} \left(J - J^{-1} \right). \tag{24}$$

The full chemical potential, including gradient regularization, is

$$\mu_s = \mu_{\mathrm{mix}}(J, T) + \mu_{\mathrm{el}}(J) - \kappa_J \frac{\partial^2 J}{\partial X^2}. \tag{25}$$

2.5 Microstructure-dependent transport

The dimensional reaction rate is

$$R(c, T, J) = k_0\, c^n \exp\left(-\frac{E_a}{R_g T} \right) a(J), \tag{26}$$

with $a(J)$ defined in Eq. (13). The effective diffusivity and mobility are

$$D_c(J) = D_0 \left(1 - \frac{\phi_{p0}}{J} \right)^{m_{\mathrm{diff}}}, \tag{27}$$

$$M_s(J) = M_0 \left(1 - \frac{\phi_{p0}}{J} \right)^{m_{\mathrm{mob}}}. \tag{28}$$

Thus the user-provided model suppresses both solute diffusion and solvent permeation in the collapsed state.

2.6 Boundary conditions

At the symmetry plane $X = 0$, the dimensional appendix imposes

$$Q_s = 0, \qquad N_c = 0, \qquad \frac{\partial T}{\partial X} = 0. \tag{29}$$

At the free surface $X = H_0$, solvent exchange is

$$Q_s = L_\mu \left(\mu_s - \mu_{\mathrm{bath}} \right), \tag{30}$$

reactant supply is

$$N_c = k_c\, (c - c_b), \tag{31}$$

and heat exchange is

$$-K_T \frac{\partial T}{\partial X} = h_T\, (T - T_\infty). \tag{32}$$

The source appendix states that the free surface is traction-free and that, in the J formulation, this is implicitly satisfied by the chemical-potential balance.

In compact nondimensional notation, the source model states

$$q\big|_1 = \mathrm{Bi}_\mu\, \mu, \qquad \mathcal{D}\, u_\xi\big|_1 = \mathrm{Bi}_c\, (1 - u), \qquad \mathcal{K}\, \theta_\xi\big|_1 = -\mathrm{Bi}_T\, \theta, \tag{33}$$

with the bath chemical potential set to $\mu_b = 0$ by convention. The fuller nondimensional appendix keeps explicit fluxes and a bath value m_b; see Eqs. (60)–(62).

3 Nondimensionalization

3.1 Scales

Length, concentration, and temperature are scaled as

$$X = H_0\,\tilde{x}, \qquad \tilde{x} \in [0,1], \tag{34}$$

$$c = c_b\,u, \tag{35}$$

$$T = T_\infty + \Delta T_*\,\theta, \qquad \Delta T_* = \frac{(-\Delta H_r)c_b}{C_*}. \tag{36}$$

The chemical potential scale is

$$\mu_s = \mu_*\,m, \qquad \mu_* = \frac{R_g T_\infty}{v_s}. \tag{37}$$

Time is scaled by the swelling/permeation time

$$\tau_s = \frac{H_0^2}{M_0\,\mu_*}. \tag{38}$$

The flux scales are

$$Q_s = \frac{H_0}{\tau_s}\,q, \qquad N_c = \frac{c_b\,H_0}{\tau_s}\,n. \tag{39}$$

Material functions are factored as

$$M_s = M_0\,\mathcal{M}(J,\theta), \qquad D_c = D_0\,\mathcal{D}(J,\theta), \tag{40}$$

$$C_{\text{eff}} = C_*\,\mathcal{C}(J), \qquad K_T = K_*\,\mathcal{K}(J). \tag{41}$$

3.2 Dimensionless reaction rate

The reference reaction rate is

$$R_0 = k_0\,c_b^n\,\exp\!\left(-\frac{E_a}{R_g T_\infty}\right), \tag{42}$$

and $R = R_0\,\widehat{R}(u,\theta,J)$ with $\widehat{R}$ given by Eq. (12). The Arrhenius parameters are

$$\varepsilon_T = \frac{\Delta T_*}{T_\infty}, \qquad \Gamma_A = \frac{E_a\,\Delta T_*}{R_g\,T_\infty^2}. \tag{43}$$

When $\varepsilon_T \ll 1$, the source notes the approximation

$$\widehat{R} \approx u^n\,a(J)\,e^{\Gamma_A\theta}. \tag{44}$$

3.3 Dimensionless chemical potential

The dimensionless chemical potential is

$$m = m_{\text{mix}}(J,\theta) + m_{\text{el}}(J) - \ell^2\,\frac{\partial^2 J}{\partial \tilde{x}^2}, \tag{45}$$

with

$$\ell^2 = \frac{\kappa_J}{\mu_*\,H_0^2}. \tag{46}$$

Neglecting the $T/T_\infty \approx 1 + \varepsilon_T \theta \approx 1$ prefactor, the source gives

$$m_{\mathrm{mix}} = \ln(1-\phi) + \phi + \chi(\theta,\phi)\,\phi^2, \qquad \phi = \frac{\phi_{p0}}{J}, \tag{47}$$

with

$$\chi(\theta,\phi) = \chi_\infty + S_\chi\,\theta + \chi_1\,\phi, \tag{48}$$

where

$$S_\chi = \Delta T_*\,\frac{d\chi_0}{dT}\bigg|_{T_\infty}. \tag{49}$$

The elastic contribution is

$$m_{\mathrm{el}} = \Omega_e\,(J - J^{-1}), \qquad \Omega_e = \frac{G\,v_s}{R_g\,T_\infty}. \tag{50}$$

3.4 Dimensionless governing equations with explicit prefactors

Dropping tildes, the appendix version of the nondimensional system is

$$\frac{\partial J}{\partial t} = \frac{\partial}{\partial x}\left(\mathcal{M}(J,\theta)\,\frac{\partial m}{\partial x}\right), \tag{51}$$

$$m = m_{\mathrm{mix}}(J,\theta) + m_{\mathrm{el}}(J) - \ell^2\,\frac{\partial^2 J}{\partial x^2}, \tag{52}$$

$$\frac{\partial (Ju)}{\partial t} + \frac{\partial}{\partial x}\left(u\,q - \delta\,\mathcal{D}(J,\theta)\,\frac{\partial u}{\partial x}\right) = -\mathrm{Da}\,J\,\widehat{R}(u,\theta,J), \tag{53}$$

where

$$q = -\mathcal{M}(J,\theta)\,\partial_x m \tag{54}$$

and

$$\delta = \frac{D_0}{M_0\,\mu_*}. \tag{55}$$

The Damköhler number is

$$\mathrm{Da} = \tau_s\,k_0\,c_b^{n-1}\,\exp\!\left(-\frac{E_a}{R_g T_\infty}\right). \tag{56}$$

The heat equation is

$$\mathcal{C}(J)\,\frac{\partial \theta}{\partial t} = \alpha\,\frac{\partial}{\partial x}\left(\mathcal{K}(J)\,\frac{\partial \theta}{\partial x}\right) + \mathrm{Da}\,J\,\widehat{R}(u,\theta,J) - \alpha\,\mathrm{Pe}_T\,\frac{\partial}{\partial x}\left[(\Theta_\infty + \theta)\,q\right], \tag{57}$$

with

$$\alpha = \frac{K_*\,\tau_s}{C_*\,H_0^2} = \frac{\tau_s}{\tau_T}, \quad \mathrm{Pe}_T = \frac{c_p^\ell\,M_0\,\mu_*}{K_*}, \quad \Theta_\infty = \frac{T_\infty}{\Delta T_*}, \tag{58}$$

where $\tau_T = C_* H_0^2/K_*$.

3.5 Dimensionless boundary conditions

At $x = 0$:

$$q = 0, \qquad u\,q - \delta\,\mathcal{D}\,u_x = 0, \qquad \theta_x = 0. \tag{59}$$

At $x = 1$:

$$q = \mathrm{Bi}_\mu\,(m - m_b), \tag{60}$$

$$u\,q - \delta\,\mathcal{D}\,u_x = \mathrm{Bi}_c\,(u - 1), \tag{61}$$

$$-\alpha\,\mathcal{K}(J)\,\theta_x = \mathrm{Bi}_T\,\theta. \tag{62}$$

The surface Biot numbers are

$$\mathrm{Bi}_\mu = \frac{L_\mu\,H_0}{M_0}, \qquad \mathrm{Bi}_c = \frac{k_c\,H_0}{M_0\,\mu_*}, \tag{63}$$

and

$$\mathrm{Bi}_T = \frac{h_T\,\tau_s}{C_*\,H_0} = \alpha\,\frac{h_T\,H_0}{K_*}. \tag{64}$$

3.6 Controlling groups

The source appendix identifies the following controlling groups: Da, α, S_χ, Γ_A, δ, ℓ, Bi_c, and Bi_T, together with working-point parameters such as ϕ_{p0} and χ_∞. The source interpretation is that oscillation requires fast positive feedback (reaction $\rightarrow$ heating $\rightarrow$ collapse) and slow negative feedback (collapse $\rightarrow$ transport suppression $\rightarrow$ cooling $\rightarrow$ re-swelling) to operate on separated timescales.

4 Well-mixed limit

The source manuscript states the uniform steady-state balances for (J_0, u_0, θ_0) as

$$\mu(J_0, \theta_0) = \mu_b, \qquad \mathrm{Bi}_c(1 - u_0) = \mathrm{Da}\,J_0\,\widehat{R}_0,$$
$$\mathrm{Bi}_T\theta_0 = \mathrm{Da}\,J_0\,\widehat{R}_0, \tag{65}$$

where $\widehat{R}_0 = \widehat{R}(u_0, J_0, \theta_0)$. These algebraic balances are used as the base state for the linear stability analysis.

The source appendix does not provide a fully separated homogeneous ODE derivation in this excerpt. It states that the $k = 0$ matrix is obtained by volume-averaging the governing equations over the slab and incorporating Robin boundary fluxes as lumped source/sink terms. A complete homogeneous ODE derivation remains a TODO for sanity checking.

5 Linear stability of the homogeneous state

5.1 Dispersion relation

Linearizing about the uniform base state with normal-mode perturbations proportional to $e^{\sigma t + ik\xi}$ gives

$$\sigma\,\widehat{\mathbf{y}} = \mathbf{A}(k)\,\widehat{\mathbf{y}}, \qquad \mathbf{A}(k) = \mathbf{A}_0 + k^2\mathbf{D}_2 + k^4\mathbf{D}_4. \tag{66}$$

The source interpretation is that $\mathbf{A}_0$ encodes reaction kinetics and surface exchange, $\mathbf{D}_2$ collects diffusive transport, and $\mathbf{D}_4$ arises from Cahn–Hilliard gradient-energy regularization.

Uniform branches are classified as stable when $\Re\sigma_1(0) < 0$, Hopf unstable when $\Re\sigma_1(0) > 0$ and $\Im\sigma_1(0) \neq 0$, and monotone unstable when $\Re\sigma_1(0) > 0$ and $\Im\sigma_1(0) = 0$.

5.2 Two instability mechanisms

For the $k = 0$ Hopf mechanism, the source manuscript identifies a positive thermal feedback path: if the Arrhenius heat–reaction coupling $\mathrm{Da}J_0 R_\theta$ exceeds surface heat loss Bi_T, a temperature perturbation increases reaction rate and heat release. The negative feedback loop is $\theta \to J \to u \to \theta$: rising temperature increases χ, drives collapse, suppresses reactant supply, and weakens the heat source. Oscillation requires the positive and negative feedback paths to operate on separated timescales.

For the finite-wavenumber spinodal mechanism, when

$$f_J \equiv \partial_J[\, m_{\mathrm{mix}} + m_{\mathrm{el}}\,] < 0, \tag{67}$$

the base state lies inside the Flory–Huggins spinodal region and the J diffusion term acts as a negative diffusivity. The k^4 term limits growth to a finite band with maximum estimated by the source as

$$k^* \approx \sqrt{-f_J/(2\ell^2)}. \tag{68}$$

The source manuscript argues that spinodal patterns do not manifest during the oscillation because the system traverses the spinodal region transiently, boundary and penetration gradients break translational symmetry, and the large-amplitude Hopf limit cycle homogenizes spatial structure each period. This is recorded here as a candidate interpretation, not as a validated repository result.

5.3 Dispersion matrix derivation

Let

$$m = f(J,\theta) - \ell^2 J_{\xi\xi}, \qquad f(J,\theta) = m_{\mathrm{mix}}(J,\theta) + m_{\mathrm{el}}(J). \tag{69}$$

Perturb about the uniform base state using

$$(\widehat{J}, \widehat{u}, \widehat{\theta})^T e^{\sigma t + ik\xi}. \tag{70}$$

Define partial derivatives at the base state:

$$f_J = \left.\frac{\partial f}{\partial J}\right|_0, \qquad f_\theta = \left.\frac{\partial f}{\partial \theta}\right|_0, \tag{71a}$$

$$R_J = \left.\frac{\partial \widehat{R}}{\partial J}\right|_0, \qquad R_u = \left.\frac{\partial \widehat{R}}{\partial u}\right|_0, \tag{71b}$$

$$R_\theta = \left.\frac{\partial \widehat{R}}{\partial \theta}\right|_0. \tag{71c}$$

The base-state transport coefficients are

$$\mathcal{M}_0 = \mathcal{M}(J_0,\theta_0), \qquad \mathcal{D}_0 = \delta\,\mathcal{D}(J_0,\theta_0), \tag{72a}$$

$$\mathcal{K}_0 = \alpha\,\mathcal{K}(J_0), \qquad \mathcal{C}_0 = \mathcal{C}(J_0). \tag{72b}$$

The $k = 0$ matrix is

$$\mathbf{A}_0 = \begin{bmatrix} a_{11} & 0 & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \tag{73}$$

with entries

$$a_{11} = -\mathrm{Bi}_\mu f_J, \qquad a_{13} = -\mathrm{Bi}_\mu f_\theta, \tag{74a}$$

$$a_{21} = \frac{u_0 \mathrm{Bi}_\mu f_J - \mathrm{Da}(\widehat{R}_0 + J_0 R_J)}{J_0}, \tag{74b}$$

$$a_{22} = \frac{-\mathrm{Bi}_c - \mathrm{Da} J_0 R_u}{J_0}, \tag{74c}$$

$$a_{23} = \frac{u_0 \mathrm{Bi}_\mu f_\theta - \mathrm{Da} J_0 R_\theta}{J_0}, \tag{74d}$$

$$a_{31} = \frac{\mathrm{Da}(\widehat{R}_0 + J_0 R_J)}{\mathcal{C}_0}, \quad a_{32} = \frac{\mathrm{Da} J_0 R_u}{\mathcal{C}_0}, \tag{74e}$$

$$a_{33} = \frac{-\mathrm{Bi}_T + \mathrm{Da} J_0 R_\theta}{\mathcal{C}_0}. \tag{74f}$$

Here $a_{12} = 0$ because the swelling equation does not depend directly on the reactant concentration.

The k^2 contribution is

$$\mathbf{D}_2 = \begin{bmatrix} -\mathcal{M}_0 f_J & 0 & -\mathcal{M}_0 f_\theta \\ 0 & -\mathcal{D}_0/J_0 & 0 \\ 0 & 0 & -\mathcal{K}_0/\mathcal{C}_0 \end{bmatrix}. \tag{75}$$

The only k^4 contribution is

$$\mathbf{D}_4 = \begin{bmatrix} -\mathcal{M}_0 \ell^2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \tag{76}$$

6 Front and barrier diagnostics

The user-provided equations contain two explicit collapse-dependent barrier factors:

$$a(J) = (1-\phi)^{m_{\mathrm{act}}}, \tag{77}$$

$$D_c(J) = D_0(1-\phi)^{m_{\mathrm{diff}}}. \tag{78}$$

The model also allows solvent mobility suppression through

$$M_s(J) = M_0(1-\phi)^{m_{\mathrm{mob}}}. \tag{79}$$

These factors are the mathematical representation of reduced reaction accessibility, reduced reactant diffusion, and reduced solvent permeation as $J \to \phi_{p0}$ and $\phi \to 1$.

The source sections ingested here do not give a final front observable for the LCST transport barrier. Candidate diagnostics to define in a follow-up include: LCST crossing position, collapsed skin thickness, reactant penetration depth, surface reactant flux, total reaction rate, and phase lag between θ, J, and reactant flux. These are TODO diagnostics, not new equations.

7 Limiting cases

7.1 Recorded limiting cases

The source derivation explicitly identifies the following limits or controls:

- $\mathrm{Pe}_T \ll 1$: enthalpy advection in the heat equation is neglected.
- $\varepsilon_T \ll 1$: the Arrhenius factor can be approximated by $e^{\Gamma_A \theta}$.
- $\mathrm{Da} = 0$ with fixed θ above LCST: the source appendix uses this as a pure Cahn–Hilliard spinodal-decomposition verification case.
- $k = 0$: homogeneous long-wave Hopf analysis.
- $k > 0$ with $f_J < 0$: finite-wavenumber spinodal analysis.

7.2 Spinodal-decomposition verification note from the source appendix

The attached source appendix describes a direct numerical simulation of pure Cahn–Hilliard dynamics with reaction switched off, $\mathrm{Da} = 0$, temperature held fixed at $\theta = 1.5$, and no-flux boundary conditions. The stated initial condition is a uniform collapsed state $J_0 = 0.30$ with $\phi = 0.50$ and small random perturbations. For that state, the source gives $f_J = -1.76 < 0$ and estimates

$$k^* = \sqrt{-f_J/(2\ell^2)} \approx 94, \tag{80}$$

with maximum growth rate

$$\sigma^* = \mathcal{M}_0 f_J^2/(4\ell^2) \approx 3850, \tag{81}$$

where $\mathcal{M}_0 = (1-\phi)^{m_{\mathrm{mob}}} = 0.50$.

This note is preserved as part of the user's preliminary appendix. The repository has not independently reproduced this numerical verification.

7.3 TODO limiting cases for sanity checking

The following limits still need to be derived or checked explicitly for this repository:

- no-reaction limit with heat equation retained;
- no-collapse or fixed-J limit;
- no-transport-degradation limit, e.g. $m_{\mathrm{diff}} = 0$ or constant diffusivity;
- no-accessibility-feedback limit, e.g. $m_{\mathrm{act}} = 0$;
- homogeneous ODE limit as a full dynamical system;
- passive thermal limit with reaction disabled and bath exchange retained;
- boundary-condition sign conventions for the dimensional and nondimensional fluxes.

8 Numerical method notes from the source appendix

This section records the numerical-method appendix as source-provided derivation context. It is not an implementation request and no PDE solver is added to this repository by EXP-001.

8.1 Mixed formulation

The source appendix introduces auxiliary variables m, q, n, and h:

$$J_t + q_x = 0, \qquad q = -\mathcal{M}(J,\theta)\, m_x, \tag{82}$$

$$m = f(J,\theta) - \ell^2 J_{xx}, \tag{83}$$

$$(Ju)_t + n_x = -\mathrm{Da}\, J\, \widehat{R}, \qquad n = u\, q - \delta\, \mathcal{D}(J,\theta)\, u_x, \tag{84}$$

$$\mathcal{C}(J)\, \theta_t + h_x = \mathrm{Da}\, J\, \widehat{R}, \qquad h = -\alpha\, \mathcal{K}(J)\, \theta_x. \tag{85}$$

The source motivation is to avoid direct fourth-order discretization by splitting the swelling–chemical-potential system into second-order equations.

8.2 Spatial discretization

For a uniform mesh of N cells,

$$\Delta x = \frac{1}{N}, \qquad x_i = \left(i - \tfrac{1}{2}\right) \Delta x, \quad i = 1, \dots, N. \tag{86}$$

Cell-centered unknowns are J_i, m_i, u_i, and θ_i; face fluxes are $q_{i+1/2}$, $n_{i+1/2}$, and $h_{i+1/2}$.

The source swelling update is

$$\frac{J_i^{n+1} - J_i^n}{\Delta t} + \frac{q_{i+1/2}^{n+1} - q_{i-1/2}^{n+1}}{\Delta x} = 0, \tag{87}$$

with

$$q_{i+1/2}^{n+1} = -\mathcal{M}_{i+1/2}^{*}\, \frac{m_{i+1}^{n+1} - m_i^{n+1}}{\Delta x}. \tag{88}$$

The face mobility is evaluated using the harmonic mean

$$\mathcal{M}_{i+1/2}^{*} = \frac{2\, \mathcal{M}_i^{*}\, \mathcal{M}_{i+1}^{*}}{\mathcal{M}_i^{*} + \mathcal{M}_{i+1}^{*}}. \tag{89}$$

The discrete chemical potential is

$$m_i^{n+1} = f(J_i^{n+1}, \theta_i^{*}) - \ell^2\, \frac{J_{i+1}^{n+1} - 2J_i^{n+1} + J_{i-1}^{n+1}}{\Delta x^2}, \tag{90}$$

with natural boundary condition $\partial_x J = 0$ imposed through ghost cells:

$$J_0^{n+1} = J_1^{n+1}, \qquad J_{N+1}^{n+1} = J_N^{n+1}. \tag{91}$$

The conserved reactant variable is $W_i = J_i u_i$:

$$\frac{W_i^{n+1} - W_i^n}{\Delta t} + \frac{n_{i+1/2}^{n+1} - n_{i-1/2}^{n+1}}{\Delta x} = -\mathrm{Da}\, J_i^{\sharp}\, \widehat{R}(u_i^{\sharp}, \theta_i^{\sharp}, J_i^{\sharp}), \tag{92}$$

followed by $u_i^{n+1} = W_i^{n+1}/J_i^{n+1}$. The flux is

$$n_{i+1/2}^{n+1} = q_{i+1/2}^{n+1}\, u_{i+1/2}^{\mathrm{up}} - \delta\, \mathcal{D}_{i+1/2}^{*}\, \frac{u_{i+1}^{n+1} - u_i^{n+1}}{\Delta x}, \tag{93}$$

with upwind value

$$u_{i+1/2}^{\mathrm{up}} = \begin{cases} u_i^{n+1}, & q_{i+1/2}^{n+1} \ge 0, \\ u_{i+1}^{n+1}, & q_{i+1/2}^{n+1} < 0. \end{cases} \tag{94}$$

The heat update is

$$\mathcal{C}_i^* \frac{\theta_i^{n+1} - \theta_i^n}{\Delta t} + \frac{h_{i+1/2}^{n+1} - h_{i-1/2}^{n+1}}{\Delta x} = \mathrm{Da}\, J_i^\sharp\, \widehat{R}(u_i^\sharp, \theta_i^\sharp, J_i^\sharp), \tag{95}$$

with

$$h_{i+1/2}^{n+1} = -\alpha\, \mathcal{K}_{i+1/2}^* \frac{\theta_{i+1}^{n+1} - \theta_i^{n+1}}{\Delta x}. \tag{96}$$

8.3 Discrete boundary conditions

At $x = 0$:

$$q_{1/2}^{n+1} = 0, \qquad n_{1/2}^{n+1} = 0, \qquad h_{1/2}^{n+1} = 0. \tag{97}$$

At $x = 1$:

$$q_{N+1/2}^{n+1} = \mathrm{Bi}_\mu\,(m_N^{n+1} - m_b), \tag{98}$$

$$n_{N+1/2}^{n+1} = \mathrm{Bi}_c\,(u_N^{n+1} - 1), \tag{99}$$

$$h_{N+1/2}^{n+1} = \mathrm{Bi}_T\,\theta_N^{n+1}. \tag{100}$$

8.4 Time integration notes

The source appendix describes a method-of-lines approach advanced by an implicit BDF integrator, plus an alternative IMEX operator-splitting strategy with stages for swelling, reactant, and temperature. For fully coupled solves, the source assembles

$$\boldsymbol{Y} = (J_1, \ldots, J_N,\ m_1, \ldots, m_N,\ W_1, \ldots, W_N,\ \theta_1, \ldots, \theta_N)^T \tag{101}$$

and solves

$$\boldsymbol{J}_{\mathrm{ac}}\, \delta \boldsymbol{Y} = -\boldsymbol{R}(\boldsymbol{Y}^{(k)}), \qquad \boldsymbol{Y}^{(k+1)} = \boldsymbol{Y}^{(k)} + \delta \boldsymbol{Y}. \tag{102}$$

The source implementation notes state that the conserved variable $W = Ju$ should be used for reactant transport, that upwinding is needed for uq, and that the fourth-order swelling equation should be handled in mixed form.

9 Open derivation issues

1. Clarify whether the canonical model is strictly first-order in reactant concentration ($n = 1$) or whether the general n-order appendix form remains allowed.

2. Reconcile notation between μ and m, between x and ξ, and between compact equations with absorbed α, δ and appendix equations with explicit α, δ.

3. Clarify the sign convention for the free-surface reactant flux, since the compact boundary condition and full flux form use different outward-flux presentations.

4. Define m_b or μ_b consistently. The compact model sets $\mu_b = 0$ by convention; the appendix keeps m_b.

5. Specify $\mathcal{C}(J)$ and $\mathcal{K}(J)$, or state explicitly that they are constant or placeholder material functions.

6. Specify whether $\mathcal{M}(J, \theta)$ keeps temperature dependence after nondimensionalization and how m_{mob} is chosen.

7. Justify the approximations $\text{Pe}_T \ll 1$ and $\varepsilon_T \ll 1$ for the intended physical regime.

8. Derive the homogeneous ODE limit explicitly rather than only the steady-state algebraic balances and $k = 0$ matrix.

9. Define front and barrier observables before using them as evidence for oscillation.

10. Treat the source appendix's numerical verification claims as unverified until reproduced under this repository's verification protocol.